

Muhammad Zaeem Ahmad

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Technical Skills

- **Languages:** JavaScript (ES6+), Python, HTML5, CSS3, TypeScript
- **Frontend Development:** React.js, Redux Toolkit, Tailwind CSS, Vite, GSAP
- **Backend and APIs:** Node.js, Express.js, Flask (Python), RESTful API Design
- **Databases and Cloud:** MongoDB, SQLite, Firebase, ImageKit, Cloudinary
- **AI and LLM Integration:** Gemini API, OpenRouter API, Large Language Model Integration
- **Security and Networking:** JWT, Cryptographic Protocols (RSA, AES, Fernet), TCP and UDP, Subnetting, Git, GitHub

Education

COMSATS University Islamabad

Bachelor of Science in Computer Science

Sahiwal, Pakistan

05/2024 – 07/2028

Relevant Experience

Freelance Software Engineer

Part-Time Developer

Sahiwal, Pakistan

11/2025 – Present

- Architected and deployed 4 production-ready web applications using the MERN stack, implementing optimized database schemas to handle application query flows.
- Secured API routing protocols with JWT, Firebase Authentication, and Role-Based Access Control (RBAC) to eliminate privilege escalation vulnerabilities.
- Integrated generative LLM interfaces (Google Gemini and OpenRouter) alongside custom error-catching algorithms to handle API quota-limits and auth issues seamlessly.
- Designed global state structures using Redux Toolkit, achieving highly performant UI rendering speeds and reducing redundant state updates.

Featured Technical Projects

AI Website Builder

Technologies: React, Tailwind CSS, Express.js, Node.js, LLM APIs

2025 – 2026

- Engineered a full-stack website creation engine enabling users to construct and edit functional web interfaces 40% faster via automated AI configuration models.
- Formulated a streamlined, modular state architecture mapping client prompts to responsive design components.

JARVIS Voice-Activated Multilingual AI Assistant

Technologies: Python, Flask, JavaScript, Gemini API, SQLite, PyAutoGUI

2025

- Constructed an autonomous, voice-controlled virtual assistant powered by a Python Flask server and real-time voice capture via the Web Speech API.
- Integrated custom pathfinding algorithms (A* Search, BFS, and DFS) for intelligent task-routing, minimizing server-side decision latency by 25%.

Lumina Studio AI Image Generation Platform

Technologies: React, TypeScript, Express.js, Google Gemini Image API

2025 – 2026

- Developed a modular image rendering web canvas integrating Gemini's image pipeline to output high-fidelity assets.
- Successfully resolved API rate-limiting bugs, introducing back-off request strategies to maintain reliable continuous uptimes under high traffic.

Professional Certifications

- Next.js App Router Fundamentals Certification
- Python Core Programming Certification